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## 12.1.1 Simple Implementation of matrix-matrix multiplication

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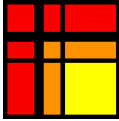
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The current coronavirus crisis hit UT-Austin on March 14, a day we spent quickly making videos for Week 11. We have not been back to the office, to create videos for Week 12, since then. We will likely add such videos as time goes on. For now, we hope that the notes suffice.

## Reading assignment

1/1 point (ungraded)



Read Unit 12.1.1 of the notes. [\[LINK\]](#)

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## Homework 12.1.1.1

1/1 point (ungraded)

In the file `Assignments/Week12/C/Gemm_IJP.c` you will find the simple implementation

```
#define alpha( i,j ) A[ (j)*ldA + i ] // map alpha( i,j ) to
array A
#define beta( i,j ) B[ (j)*ldB + i ] // map beta( i,j ) to
array B
#define gamma( i,j ) C[ (j)*ldC + i ] // map gamma( i,j ) to
array C

void MyGemm( int m, int n, int k, double *A, int ldA,
            double *B, int ldB, double *C, int ldC )
{
    for ( int i=0; i<m; i++ )
        for ( int j=0; j<n; j++ )
            for ( int p=0; p<k; p++ )
                gamma( i,j ) += alpha( i,p ) * beta( p,j );
}
```

that computes  $C := AB + C$ . In the directory `Assignments/Week12/C` execute

make IJP

to compile, link, and execute it. You can view the performance attained

on your computer with the Matlab Live Script in [Assignments/Week12/C/data/Plot\\_IJP.mlx](#) (Alternatively, read and execute [Assignments/Week12/C/data/Plot\\_IJP\\_m.m>.](#))

☒ done



Solution:

For a complete solution, see the text below Homework 12.1.1.1 in [the notes](#).

There is a video in the notes.

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Answers are displayed within the problem

Homework 12.1.1.2

1/1 point (ungraded)  
The IJP ordering is one possible ordering of the loops. How many distinct reorderings of those loops are there?

☐  $2^3 = 8$

☒  $3! = 6$

☐ 3



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Homework 12.1.1.3

1/1 point (ungraded)  
In directory `Assignments/Week12/C` make copies of `Gemm_IJP.c` into files with names that reflect the different loop orderings ( `Gemm_IPJ.c` , etc.). Next, make the necessary changes to the loops in each file to reflect the ordering encoded in its name. Test the implementations by executing

```
make IPJ
make JIP
...
```

for each of the implementations and view the resulting performance by making the indicated changes to the Live Script in [Assignments/Week12/C/data/Plot\\_All\\_Orderings.mlx](#) (Alternatively, use the script in [Assignments/Week12/C/data/Plot\\_All\\_Orderings\\_m.m](#)). If you have implemented them all, you can test them all by executing

make All\_Orderings

☒ done

 done



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### Homework 12.1.1.4

1/1 point (ungraded)

In directory `Assignments/Week12/C/` execute

make JPI

and view the results with the Live Script in `Assignments/Week12/C/data/Plot_Openers.mlx` . (This may take a little while, since the Makefile now specifies that the largest problem to be executed is m=n=k=1500.)

Next, change that Live Script to also show the performance of the reference implementation provided by BLIS: Change

% Optionally show the reference implementation performance data  
if ( 0 )

to

% Optionally show the reference implementation performance data  
if ( 1 )

and rerun the Live Script. This adds a plot to the graph for the reference implementation.

What do you observe? Now are you happy with the improvements you made by reordering the loops>

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